

Самостоятельная работа 2.7

Решение линейного уравнения

первого порядка.

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В1.

$$1) y' + 3 \frac{y}{x} + 9x = 0$$

$$y' + \frac{3y}{x} = -9x$$

$$y = uv \quad y' = u'v + uv'$$

$$u'v + \frac{3uv}{x} = -9x$$

$$v' + \frac{3v}{x} = 0$$

$$v = -\frac{3}{x}$$

$$\frac{dv}{v} = -\frac{3}{x} dx$$

$$\ln v = -3 \ln x$$

$$v = x^{-3}$$

$$u' x^{-3} = -9x$$

$$u' = -9x^4$$

$$u = -9 \int x^4 dx = -9 \cdot \frac{x^5}{5} + C$$

$$y = \left(-\frac{9}{5} x^5 + C \right) x^{-3}$$

$$y = -\frac{9}{5} x^2 + C x^{-3}$$

$$y = -\frac{9}{5} x^2 + \frac{C}{x^3}$$

$$\sqrt{2}$$

$$y' + \frac{y}{x} = 2x$$

$$y(1) = 2$$

$$y = uv$$

$$y' = u'v + u v'$$

$$u'v + u u' + \frac{uv}{x} = 2x$$

$$v' + \frac{v}{x} = 0$$

$$v = -\frac{v}{x}$$

$$\frac{dv}{v} = -\frac{1}{x} dx$$

$$\ln v = -\ln x$$

$$v = x^{-1}$$

$$u' x^{-1} = 2x$$

$$u = 2x^2$$

$$u = 2 \int x^2 dx = 2 \cdot \frac{x^3}{3} + C$$

$$u = \frac{2}{3} x^3 + C$$

$$y = \left(\frac{2}{3} x^3 + C \right) x^{-1} = \frac{2}{3} x^2 + \frac{C}{x}$$

$$y(1) = 2 \therefore 2 = \frac{2}{3} + C$$

$$C = 2 - \frac{2}{3} = \frac{4}{3}$$

$$y = \frac{2}{3} x^2 + \frac{4}{3x}$$